

2082

Bachelor of Science in Computer Science and Information Technology

Course Title: **Operating Systems**

Code No: **CSC264** Semester: **4th Semester**

Full Marks: 60 Pass Marks: 24 Time: **quantum value and context switching time affect each other, in a round-robin scheduling algorithm.9.Five batch jobs A, B, C, D and E, with estimated running times 10, 6, 2, 4 and 8 minutes respectively, arrived at the same time. Determine the average turnaround time for the scheduling algorithms FCFS and SJF.10.Given the page reference string: 1, 2, 3, 4, 2, 1, 5, 6, 2, 7, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6, compute the numbers of page faults for FIFO page replacement algorithm.11.What is Belady's anomaly? For the decimal virtual addresses, 20000, 32168, 60000, compute the virtual page number and offset for a 4-KB page and for an 8 KB page.12.Differentiate between**a) Preemptive and non-preemptive scheduling.b) Physical address and logical address.

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. Explain the differences between deadlock, livelock and starvation. A system has four processes and five allocated resources. Given the following current allocation and maximum needs, find the smallest value of x for which this is a safe state.
Process Allocated Maximum Available
A 10 21 11 12 13 00 x 11 B 20 11 02 22 10 C 11 01 02 13 10 D 11 11 01 12 21
2. How the files are implemented using linked list allocation? How many disk operations are needed to fetch the i-node for a file with the path name /usr/ast/courses/os/handout.t? Assume that the i-node for the root directory is in memory, but nothing else along the path is in memory. Also assume that all directories fit in one disk block.
3. If a disk has double interleaving, does it also need cylinder skew in order to avoid missing data when making a track-to-track seek? Justify your answer. Consider a disk queue with requests for I/O to blocks on cylinders 24, 90, 133, 43, 188. The head is initially at cylinder number 100 and the cylinders are numbered from 0 to 199. Calculate the total head movement for servicing the request using FCFS, SSTF and CSCAN.

Section B

Attempt any Eight questions[8×5=40]

4. List and describe two main functions of operating system.
5. Define segmentation. Why do we need multiple queues in process scheduling?
6. What do you mean by open source operating systems? Why do you think the designers of Linux made it impossible for a process to send a signal to another process that is not in its process group?
7. During implementing mutual exclusion, how can you solve the problem of busy wait? Explain.

-
8. What is race condition? Explain how time quantum value and context switching time affect each other, in a round-robin scheduling algorithm.
 9. Five batch jobs A, B, C, D and E, with estimated running times 10, 6, 2, 4 and 8 minutes respectively, arrived at the same time. Determine the average turnaround time for the scheduling algorithms FCFS and SJF.
 10. Given the page reference string: 1, 2, 3, 4, 2, 1, 5, 6, 2, 7, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6, compute the numbers of page faults for FIFO page replacement algorithm.
 11. What is Belady's anomaly? For the decimal virtual addresses, 20000, 32168, 60000, compute the virtual page number and offset for a 4-KB page and for an 8 KB page.
 12. Differentiate between a) Preemptive and non-preemptive scheduling. b) Physical address and logical address.